

ANTI-EPILEPTIC DRUGS

BY

PROF. DR.RIFFAT FAROOQUI

DUHS

EPILEPSY

**ABNORMAL DISCHARGE OF
NEURONAL CELLS WITH
ASSOCIATED MOTOR SENSORY OR
BEHAVIORAL CHANGES**

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CAUSES OF EPILEPSY

- ☐ TRAUMA
- ☐ DRUGS
- ☐ WITHDRAWAL FROM DEPRESSANTS
- ☐ TUMOR
- ☐ HIGH FEVER
- ☐ HYPOGLYCEMIA
- ☐ EXTREME ACIDOSIS
- ☐ EXTREME ALKALOSIS
- ☐ IDIOPATHIC



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CLASSIFICATION OF EPILEPTIC SEIZURES

I. PARTIAL (FOCAL) SEIZURES

A. SIMPLE PARTIAL SEIZURES

B. COMPLEX PARTIAL SEIZURES

II. GENERALIZED SEIZURES

A. TONIC-CLONIC SEIZURES

B. ABSENCE SEIZURES

C. TONIC SEIZURES

D. CLONIC SEIZURES

E. ATONIC SEIZURES

F. MYOCLONIC SEIZURES

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A. SIMPLE PARTIAL SEIZURES (JACKSONIAN)

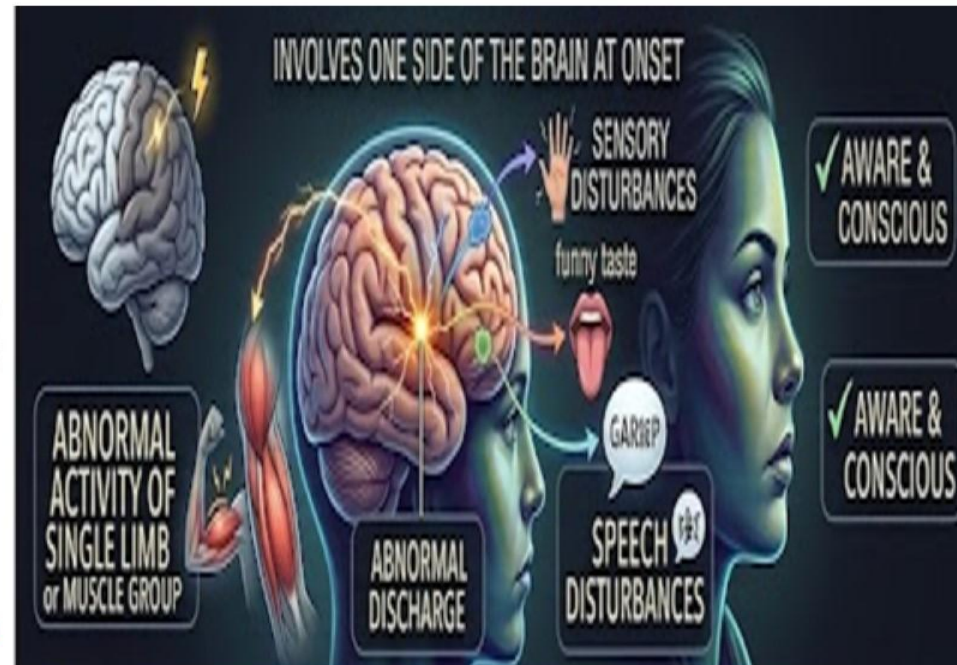
- INVOLVES ONE SIDE OF THE BRAIN AT ONSET.
- ABNORMAL DISCHARGE CONFINED TO SINGLE LOCUS WITH MOTOR, SENSORY OR SPEECH DISTURBANCES.
- ABNORMAL ACTIVITY OF SINGLE LIMB OR MUSCLE GROUP.
- NO ALTERATION OF CONSCIOUSNESS.

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SIMPLE PARTIAL SEIZURES

- ABNORMAL DISCHARGE CONFINED TO SINGLE LOCUS WITH MOTOR, SENSORY.
- ABNORMAL ACTIVITY OF SINGLE LIMB OR MUSCLE GROUP.
- NO ALTERATION OF CONSCIOUSNESS.

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B. COMPLEX PARTIAL SEIZURES (TEMPORAL LOBE EPILEPSY OR PSYCHOMOTOR SEIZURES)

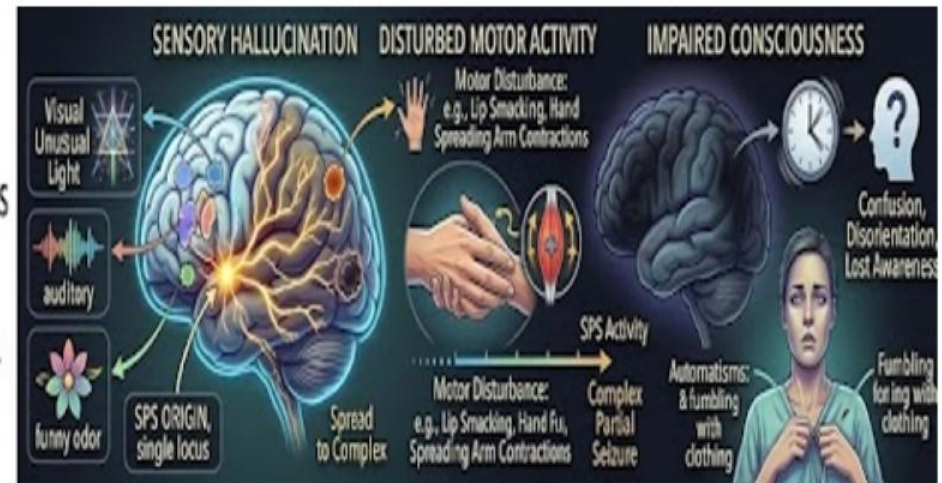
- ❑ CONSCIOUSNESS IS IMPAIRED OR LOST
- ❑ MOTOR ACTIVITY IS DISTURBED
- ❑ SENSORY HALLUCINATIONS
- ❑ SPS ACTIVITY MAY SPREAD TO BECOME COMPLEX

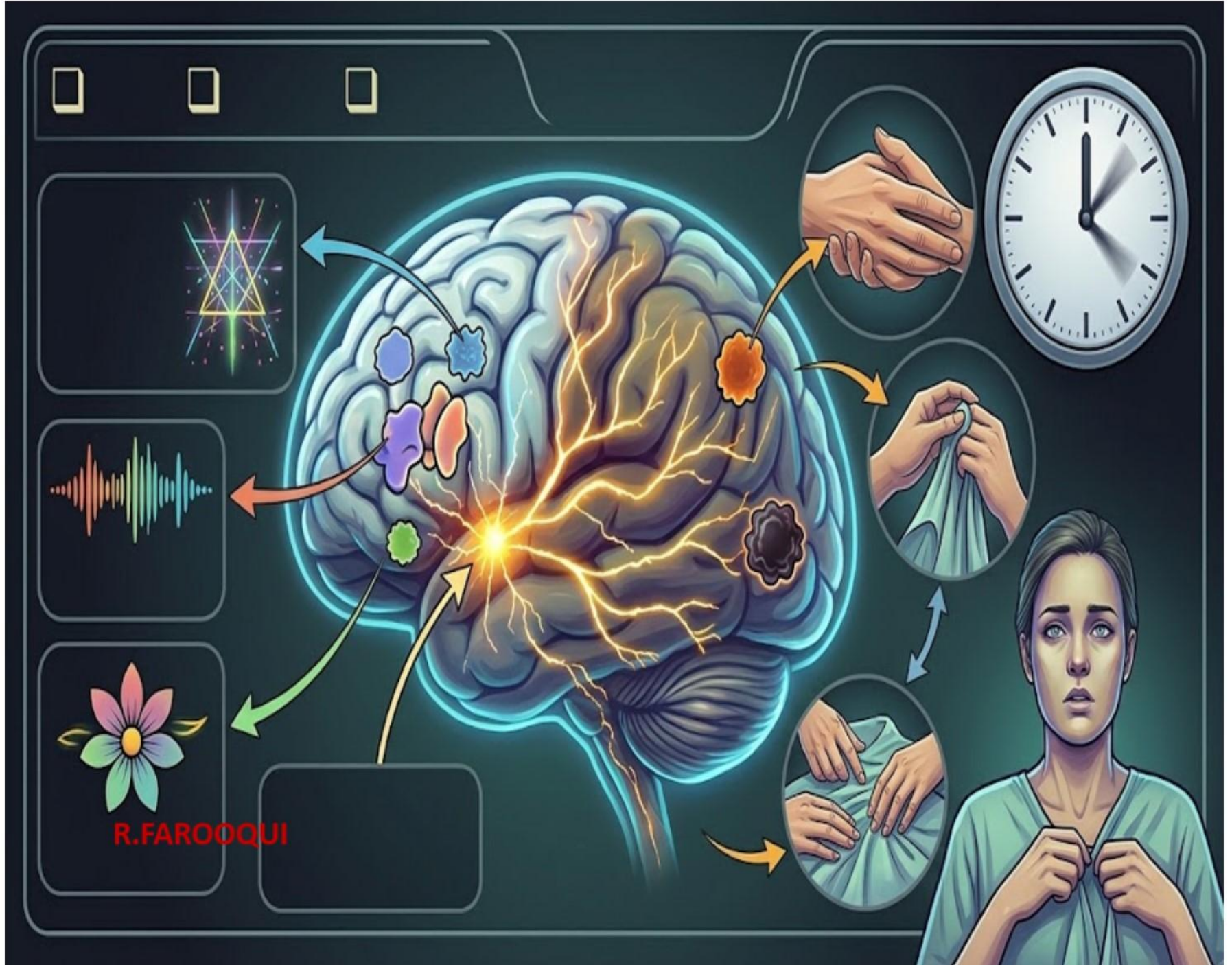
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COMPLEX PARTIAL SEIZURES SIMPLE PARTIAL SEIZURES

- ❑ CONSCIOUSNESS IS IMPAIRED OR LOST
 - ❑ MOTOR ACTIVITY IS DISTURBED
 - ❑ SENSORY HALLUCINATIONS
 - ❑ SPS ACTIVITY MAY SPREAD TO BECOME COMPLEX
- ABNORMAL DISCHARGE CONFINED TO SINGLE LOCUS WITH MOTOR, SENSORY.
 - ABNORMAL ACTIVITY OF SINGLE LIMB OR GROUP.
 - NO ALTERATION OF CONSCIOUSNESS.

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II. GENERALIZED SEIZURES

- **BOTH HEMISPHERES ARE WIDELY INVOLVED**
- **MAY BE CONVULSIVE OR NONCONVULSIVE**
- **IMMEDIATE LOSS OF CONCIUSNESS**

A.TONIC-CLONIC SEIZURES

MAJOR CONVULSIONS, USUALLY WITH TWO PHASES:

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1) TONIC PHASE

2) CLONIC PHASE

TONIC-CLONIC SEIZURES (GRAND MAL)

EEG
LOC



EEG Spike-Wave

Tonic Phase:
Rigid muscles,
loss of consciousness

Clonic Phase:
Rhythmic jerking,
convulsive
movements



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TONIC SEIZURES



Sudden muscle
stiffening; fixed
position



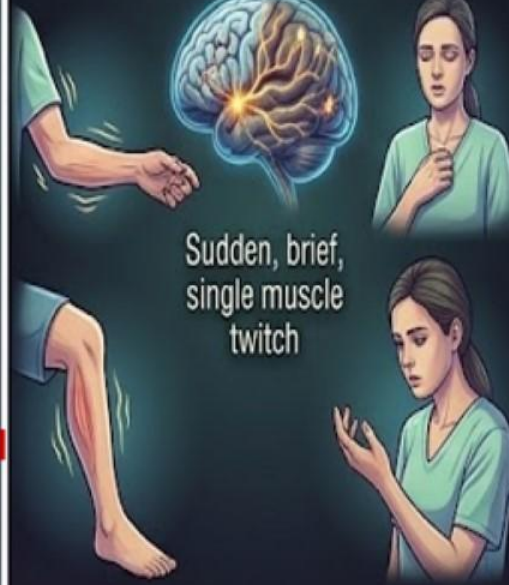
CLONIC SEIZURES



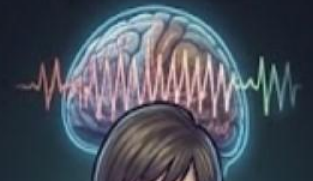
MYOCLONIC SEIZURES



Sudden, brief,
single muscle
twitch



ABSENCE SEIZURE (PETIT MAL)



Brief loss of consciousness;
staring blankly;
no convulsions



TONIC PHASE

- **SUSTAINED POWERFUL MUSCLE CONTRACTION**

(INVOLVING ALL BODY MUSCULATURE)

- **ARRESTS VENTILATION**

CLONIC PHASE:

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ALTERNATING CONTRACTION AND RELAXATION

B. TONIC SEIZURES

- **Increase Tone In Extension Muscles**

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Duration: Less Than 60 Sec

C. ATONIC SEIZURES (*ATYPICAL*)

- **Loss Of Postural Tone**
- **May Lose Consciousness**

D.MYOCLONIC SEIZURES:

- ☐ Short Episodes Of Muscle Contraction That Recur For Several Min
- ☐ Occur After Awakening
- ☐ Brief Jerks Of The Limb

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E.ABSENCE SEIZURES

- ☐ No Convulsion
- ☐ Brief, Abrupt And Self-Limiting Loss Of Consciousness

TREATMENT OF SEIZURES

GOALS:

- ☐ BLOCK REPETITIVE NEURONAL FIRING.
- ☐ BLOCK SYNCHRONIZATION OF NEURONAL DISCHARGES.
- ☐ BLOCK PROPAGATION OF SEIZURE.
- ☐ MINIMIZE SIDE EFFECTS WITH THE SIMPLEST DRUG REGIMEN.
- ☐ MONOTHERAPY IS RECOMMENDED IN MOST CASES

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TREATMENT STRATEGIES:

- **MODIFICATION OF ION CONDUCTANCES.**
- **INCREASE INHIBITORY (GABAERGIC) TRANSMISSION.**
- **DECREASE EXCITATORY (GLUTAMATERGIC) ACTIVITY**

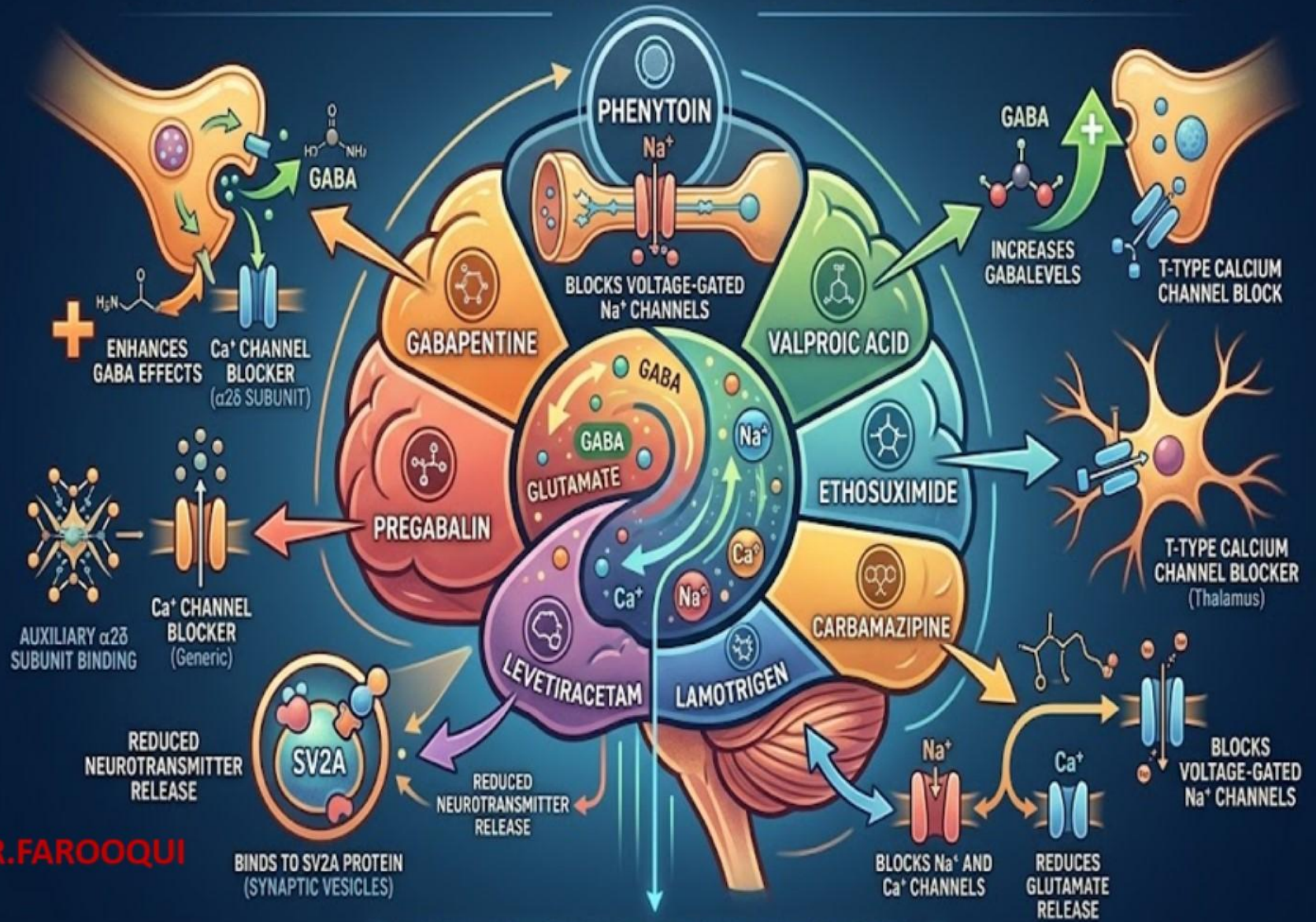
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Classification of Anticonvulsants

Action on Ion Channels	Enhance GABA Transmission	Inhibit EAA Transmission
Na⁺: Phenytoin, Carbamazepine, Lamotrigine Topiramate Valproic acid Ca⁺⁺: Ethosuximide Valproic acid	Benzodiazepines (diazepam, clonazepam) Barbiturates (phenobarbital) Valproic acid Gabapentin Vigabatrin Topiramate Felbamate	Felbamate Topiramate R.FAROOQUI

AVATAR

AVATAR MECHANISMS OF ACTION OF ANTI-EPILEPTIC DRUGS (AEDS)



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COMBINED SYNERGISTIC MECHANISMS FOR SEIZURE CONTROL

PHENYTOIN (DIPHENYLHYDANTOIN)

- ❑ OLDEST NONSEDATIVE ANTIEPILEPTIC DRUG
- ❑ BLOCKS VOLTAGE GATED SODIUM CHANNELS
- ❑ PROLONGS THE INACTIVATED STATE OF VOLTAGE SENSITIVE NEURONAL Na^+ CHANNEL AND REDUCES THE NEURONAL EXCITABILITY
- ❑ FOSPHENYTOIN, A MORE SOLUBLE PRODRUG IS USED FOR PARENTERAL USE

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ADVERSE EFFECTS OF PHENYTOIN: THE MNEMONIC AVATAR "PHENY"

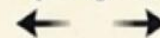
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ADVERSE EFFECTS OF PHENYTOIN: THE MNEMONIC AVATAR "PHENY"

Ataxia /
Cerebellar
Dysfunction



Nystagmus



Hirsutism



Fetal Hydantoin
Syndrome

- Microcephaly
- Cleft Palate
- Heart Defects



Peripheral Neuropathy,
numbness/sensation loss

ADRs PHENYTOIN

- ☐ ATAXIA
- ☐ NYSTAGMUS
- ☐ DIPLOPIA
- ☐ COGNITIVE IMPAIRMENT
- ☐ HIRSUTISM
- ☐ GINGIVAL HYPERPLASIA
- ☐ DOSE-DEPENDENT ZERO ORDER KINETICS (Non-Linear)
- ☐ FETAL HYDANTOIN SYNDROME
- ☐ PERIPHERAL NEUROPATHY
- ☐ CYP450 INDUCER

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CLINICAL USES OF PHENYTOIN

1. GENERALIZED TONIC-CLONIC, SIMPLE AND COMPLEX PARTIAL SEIZURES.(DRUG OF CHOICE)

- DOSE: 100 MG BD, MAXIMUM 400 MG/DAY

2. STATUS EPILEPTICUS: OCCASIONALLY USED BY SLOW I.V. INJECTION.

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3. MIGRAINE

CARBAMAZIPINE

- ❑ MECHANISM OF ACTION: BLOCKS SODIUM CHANNELS INHIBITS HIGH FREQUENCY REPETITIVE FIRING.
- ❑ ENZYME INDUCER

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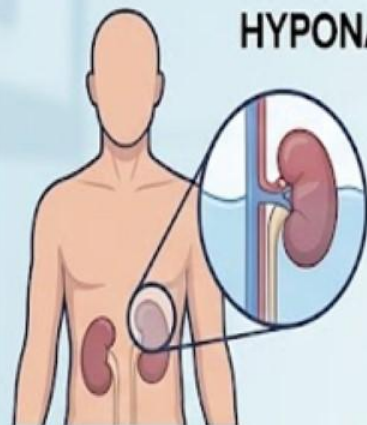
CARBAMAZEPINE ADVERSE EFFECTS

- ☐ **DIPLOPIA**
- ☐ **BLOOD DYSCRASIAS**
- ☐ **COGNITIVE DYSFUNCTION**
- ☐ **STEVENS JOHNSON SYNDROME**
- ☐ **WATER RETENTION AND HYPONATREMIA IN THE ELDERLY AS IT ENHANCES ADH ACTION.**
- ☐ **TERATOGENIC (CRANIOFACIAL ANOMALIES AND SPINA BIFIDA)**
- ☐ **DDI**

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ADVERSE EFFECTS OF CARBAMAZEPINE

HYPONATREMIA (SIADH)



Excessive Water Retention

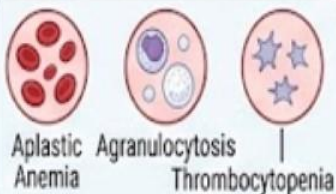


Low Plasma Sodium (<135 mEq/L)

Nausea, fatigue, seizures (if severe)

Nausea, fatigue, seizures (if severe)

BLOOD DYSCRASIAS (Bone Marrow Suppression)



Must monitor CBC
(Complete Blood Count)

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TERATOGENICITY



Neural Tube Defects
(Spina Bifida)

Cleft Lip/Palate

Cardiovascular
malformations



Contraindicated in pregnancy
without specialist review

STEVENS-JOHNSON SYNDROME (SJS)



Medical
Emergency

Fever, influenza-like symptoms,
followed by a painful red/purple rash
and mucosal involvement
(HINT: Check for blister formation)

Medical Emergency

DIPLOPIA



Double vision, blurred
vision (dose-related)

COGNITIVE DYSFUNCTION



Memory impairment, slowed
thinking, difficulty concentrating

Memory impairment, slowed
thinking, (dose-related)

Requires regular monitoring:
Sodium, CBC, HLA-B*1502 testing

CARBAMAZEPINE THERAPEUTIC USES

❑ DRUG OF CHOICE IN

❖ PARTIAL SEIZURES

❖ GENERALIZED TONIC CLONIC SEIZURES

❖ TRIGEMINAL NEURALGIA

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❑ MANIC DEPRESSIVE ILLNESS AND
ACUTE MANIA – AS AN ALTERNATIVE
TO LITHIUM

VALPROIC ACID/SODIUM VALPORATE

- ❑ FREE ACID FULLY IONIZED AT BODY PH, THUS ACTIVE FORM IS VALPROATE ION.
- ❑ MECHANISM OF ACTION
 - ❖ INCREASES LEVELS OF GABA IN BRAIN
 - ❖ BLOCKS VOLTAGE-DEPENDENT SODIUM CHANNELS
 - ❖ BLOCKS T-TYPE CALCIUM CURRENTS
 - ❖ INHIBITS THE GABA-TRANSAMINASE
 - ❖ CYP₄₅₀ INHIBITOR
 - ❖ DIVALPROEX: CONVERTED TO VALPORATE HAS IMPROVED GIT TOLERANCE

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VALPROIC ACID ADRS

- ☐ **HEPATOTOXICITY**
- ☐ **ALOPECIA**
- ☐ **WEIGHT GAIN**
- ☐ **TERATOGENIC (SPINA BIFIDA)**
- ☐ **DDI (INHIBITOR OF P₄₅₀)**

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THERAPEUTIC USES

- ❑ HIGHLY EFFECTIVE IN ABSENCE SEIZURE
- ❑ ALTERNATIVE /ADJUVANT DRUG FOR GTCS, SPS, CPS
- ❑ MYOCLONIC AND ATONIC SEIZURES—DRUG OF CHOICE
- ❑ MANIA AND BIPOLAR ILLNESS: AS ALTERNATIVE TO LITHIUM

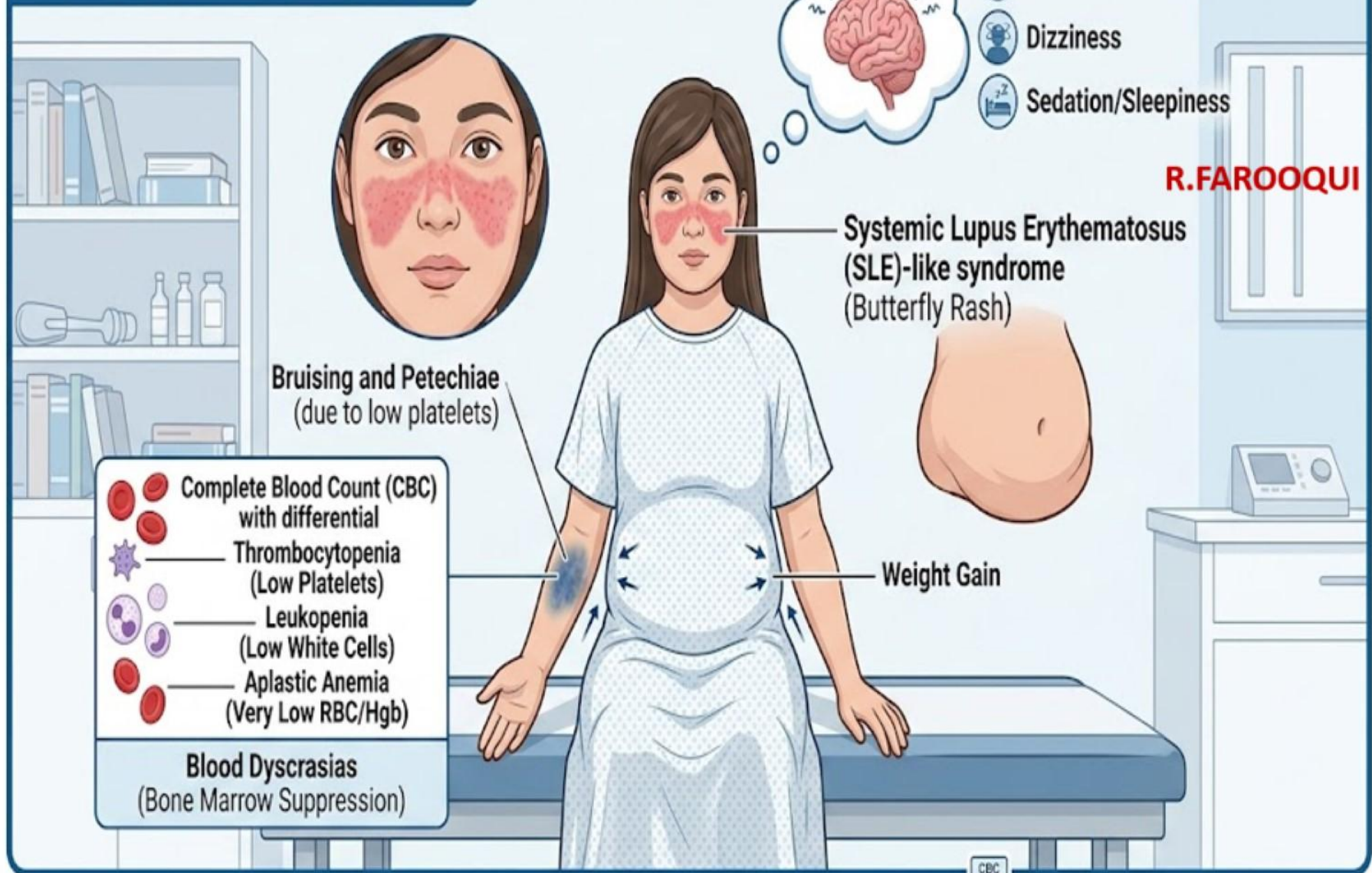
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ETHOSUXIMIDE

- ❑ BLOCK T –TYPE CALCIUM CHANNELS
- ❑ ONLY EFFECTIVE IN ABSENCE SEIZURES
- ❑ ADRs:
 - ❖ WEIGHT GAIN
 - ❖ SLE
 - ❖ BLOOD DYSCRASIA

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ETHOSUXIMIDE



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ADVERSE EFFECTS OF ETHOSUXIMIDE: 'ESME'S MNEMONIC AVATAR'



Regular CBC Monitoring Required

LAMOTRIGEN

- ❑ BLOCK SODIUM CHANNELS AND HIGH VOLTAGE DEPENDENT CALCIUM CHANNELS
- ❑ P450 INDUCERS INCREASES ITS METABOLISM
- ❑ DIVALPROEX(DDI) AN INHIBITOR OF P450 INHIBIT ITS METABOLISM CAUSES HIGH LAMOTRIGEN LEVEL.
- ❑ LAMOTRIGEN +DIVALPROEX = SERIOUS RASH
- ❑ USES
 - ❖ SIMPLE PARTIAL SEIZURE---DRUG OF FIRST CHOICE
 - ❖ GENERALIZED SEIZURES
 - ❖ ABSENCE SEIZURES
 - ❖ LENNOX GASTAUT SYNDROME
 - ❖ BIPOLAR DISORDERS

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ADRs

- ❖ STEVEN JOHNSON SYNDROME

1. Key Seizure Types:

Tonic Seizure



Atypical Absence Seizure



Drop Seizure (Atonic/Tonic-Atonic)



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LENNOX-GASTAUT SYNDROME (LGS)

CORE FEATURES & DIAGNOSTIC TRIAD



A. SEIZURE TYPES

- Tonic seizures (most common)
- Atypical absence seizures
- Atonic (drop) seizures



B. EEG FINDINGS

- Slow Spike-and-Wave (SSW) discharges (<2.5 Hz)
- Typically generalized and during sleep



C. INTELLECTUAL DEVELOPMENT

- Progressive intellectual disability and cognitive impairment
- Behavior problems (e.g., hyperactivity, autism, aggression)

D. ADDITIONAL FINDINGS

- Common onset before age 8 (typically 3-5 years)
- Severe and treatment-resistant (intractable) epilepsy
- Association with varying etiologies (genetic, structural, or metabolic)
- High risk of injuries (from drop attacks)

LEVETIRACETAM

- BINDS TO SYNAPTIC VESICLE PROTEIN (SV2A)
- EXCRETED IN URINE MOSTLY UNCHANGED- NO DDI
- CAUSE MOOD ALTERATION
- USES (AS ADJUNCT)
 - ☐ SPS
 - ☐ GENERALISED TONIC – CLONIC SEIZURES
 - ☐ MYOCLONIC SEIZURE

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TOPIRAMATE

- ❑ BLOCK SODIUM CHANNELS
- ❑ BLOCK L-TYPE CALCIUM CHANNELS
- ❑ BLOCK GLUTAMATE RECEPTORS
- ❑ CARBONIC ANHDRASE INHIBITOR
- ❑ USES:
 - ❖ SIMPLE PARTIAL SEIZURE
 - ❖ GENERALISED SEIZURE
 - ❖ MIGRAINE
- ❑ ADRs
 - ❖ WEIGHT LOSS
 - ❖ OLIGOHDROSIS
 - ❖ RENAL STONES
 - ❖ PARAESTHESA
 - ❖ GLUCOMA
 - ❖ SOMNOLANCE

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GABAPENTINE

- ❑ INHIBITS VOLTAGE GATED CALCIUM (alpha-2-delta subunit) CHANNELS
- ❑ INCREASES GABA RELEASE
- ❑ USES
 - ❖ POST-HERPETIC NEURALGIA
 - ❖ DIABETIC NEURALGIA
 - ❖ MIGRAINE
 - ❖ SIMPLE PARTIAL SEIZURE AS AN ALTERNATIVE
- ❑ GOOD CHOICE IN ELDERLY (FEW DDI AND ADRs)
- ❑ NONLINEAR PHARMACOKINETICS

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PREGABALIN

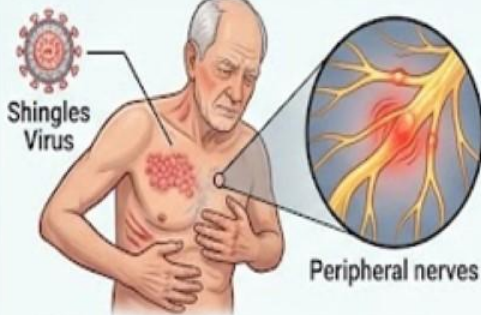
- ❑ INHIBIT VOLTAGE GATED CALCIUM (ALFA 2 DELTA SITE) CHANNELS, INHIBITING EXCITATORY NEUROTRANSMITTER RELEASE.
- ❑ USES
 - ❖ POST-HERPETIC NEURALGIA
 - ❖ SIMPLE PARTIAL SEIZURES AS AN ALTERNATIVE
 - ❖ DIABETIC PN
 - ❖ FIBROMYALGIA
- ❑ 90% EXCRETED BY KIDNEY , NO SIGNIFICANT EFFECT ON METABOLISM , FEW DDI

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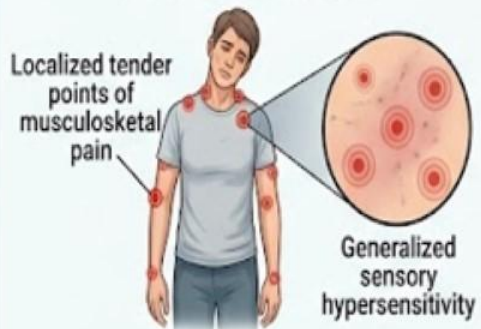
CLINICAL USES OF GABAPENTIN & PREGABALIN

POST HERPETIC NEURALGIA

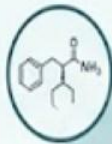


Relieves chronic nerve pain from Varicella-Zoster infection

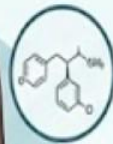
FIBROMYALGIA



Treats widespread musculoskeletal pain and associated fatigue



Gabapentin



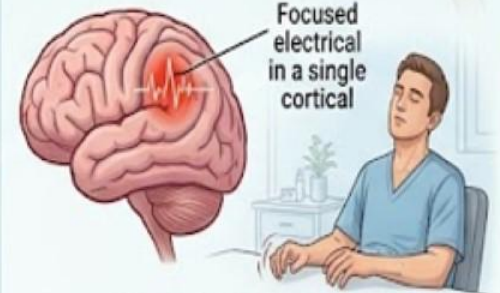
Pregabalin

DIABETIC PERIPHERAL NEUROPATHY



Alleviates neuropathic pain in the extremities (Feet & Hands)

SIMPLEE PARTIAL SEIZURES



Used as adjunctive therapy for localized (partial) seizures without consciousness impairment

MIGRAINE



Effective for prophylaxis and acute treatment of debilitating headaches

BENZODIAZEPINES

- ❑ INTERACT WITH SPECIFIC RECEPTORS ON THE GABA RECEPTORS
- ❑ INCREASES FREQUENCY OF OPENING OF CHLORIDE ION CHANNELS
- ❑ FACILITATE INHIBITORY EFFECT OF GABA
- ❑ USES: I/V DIAZEPAM OR LORAZEPAM IN STATUS EPILEPTICUS

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PHENOBARBITAL

- ❑ INTERACT WITH A DIFFERENT SITE ON CHLORIDE ION CHANNEL
- ❑ INCREASES DURATION OF CHLORIDE ION CHANNEL OPENING
- ❑ ENHANCE INHIBITORY ACTION OF GABA

❑ USES:

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- ❖ SIMPLE PARTIAL SEIZURE AS AN ALTERNATIVE
- ❖ INFANTILE SEIZURE (DRUG OF CHOICE)
- ❖ STATUS EPILEPTICUS IN CHILDREN

FELBAMATE

❑ BLOCK NMDA RECEPTORS/BLOCK SODIUM CHANNELS/BLOCK CALCIUM CHANNELS/POTENTIATE GABA ACTION

❑ CYP2C19 AND INDUCER CYP3A4INHIBITOR

❑ USES:

RESERVED FOR REFRACTORY EPILEPSIES (LENNOX-GASTAUT SYNDROME) DUE TO SIDE EFFECTS AND DDI

❑ ADRs

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❖ APLASTIC ANEMIA

❖ HEPATIC FAILURE

TIAGIBINE:

- ❑ DEPRESSING GABA TRANSPORTER GAT-1 ,
INCREASES GABA LEVELS

VIGABATRIN:

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- ❑ IRREVERSIBLY INACTIVATE GABA
AMINOTRANSFERASE (GABA-T)
- ❑ VISUAL FIELD LOSS

MECHANISM F ACTIONS OF ANTIEPILEPTICS

SODIUM CHANNEL BLOCKERS

- ❖ PHENYTOIN
- ❖ CARBAMAZIPINEVALPROIC ACID
- ❖ TOPIRAMATE
- ❖ LAMOTRIGEN
- ❖ FELBAMATE

CALCIUM CHANNEL BLOCKERS

- ❖ VALPROIC ACID
- ❖ GABAPENTIN
- ❖ PREGABALIN
- ❖ ETHOSUXIMIDE
- ❖ LAMOTRIGEN
- ❖ FELBAMATE

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POTENTIATION OF GABA

- ❖ DIAZEPAM
- ❖ PHENOBARBITAL
- ❖ GABAPENTIN
- ❖ TOPIRAMATE
- ❖ VALPROIC ACIS

NMDA BLOCKERS

- ❖ TOPIRAMATE
- ❖ VALPROIC ACID
- ❖ FELBAMATE

ADVERSE EFFECTS

☐ SPINA BIFIDA

❖ VALPROIC ACID

❖ CARBAMAZIPINE

☐ FETAL HYDANTOIN SYNDROME

❖ PHENYTOIN

☐ HEPATOTOXICITY

❖ VALPROIC AVD

❖ CARBAMAZIPINE

❖ PHENYTOIN

❖ FELBAMATE

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ADVERSE EFFECTS

☐ BLOOD DYSCRASIA

❖ CARBAMAZIPINE

❖ ETHOSUXIMIDE

❖ FELBAMATE

☐ STEVEN JOHNSON SYNDROME

❖ CARBAMAZIPINE

❖ LAMOTRIGEN

❖ SLE

ETHOSUXIMIDE

OSTEOPOROSIS/OSTEOMALACIA

❖ PHENYTOIN

❖ PHENOBARBITAL

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ADVERSE EFFECTS

☐ WEIGHT GAIN

❖ ETHOSUXIMIDE

❖ FELBAMATE

❖ VALPROIC ACID

☐ WEIGHT LOSS

❖ TOPIRAMATE

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ANTIPILEPTICS QUESTION
BRAIN WARM UP QUIZ

1. SELECT DRUG THAT SIGNIFICANTLY INCREASES PLASMA LEVEL OF LAMOTRIGEN?
2. SELECT THE MOA OF ETHOSUXIMIDE.
3. SELECT DRUG THAT POSSES NON-LINEAR KINETICS.
4. SELECT DRUG THAT CAUSES HYPONATREMIA .
5. SELECT MOA OF TOPIRAMATE. R.FAROOQUI
6. SELECT THE CLINICAL INDICATION OF CARBAMAZIPINE.
7. SELECT THE MOA OF PREGABALIN.

	OPTIONS	WRITE QUESTION NUMBER
A	TRIGEMINAL NEURALGIA	
B	BLOCK T-TYPE CALCIUM CHANNELS	
C	CARBAMAZIPINE	
D	BINDS TO ALFA -2-DELTA SUBUNIT OF CALCIUM CHANNEL DECREASES NEURONAL EXCITABILITY	
E	PHENYTOIN R.FAROOQUI	
F	DIVALPROEX	
G	BLOCK SODIUM CHANNEL, L-TYPE CALCIUM CHANNEL, BLOCK NMDA RECEPTO	

THANK YOU